

IQRA UNIVERSITY

Faculty of Engineering Sciences and Technology

# Human Computer Interaction

## Complex Computing Problem (CCP) Report



### Project Title:

## Scribbly: Stationery E-Commerce Website

#### Group Members:

Syed Imran Murtaza (65196)  
Muhammad Shehreyar (65193)  
Ahmed Abdul Qayyum (61343)

#### Instructor:

Ma'am Zuha Soomro

#### Website Live Preview:

<https://ctrlaltimran.github.io/Scribbly-HCI-Project-/index.html>

#### Project Files/Repository:

<https://github.com/ctrlaltimran/Scribbly-HCI-Project->

# 1. Introduction and Project Overview

## Project Background

Stationery websites often need to serve users who want to make quick choices from visually similar products. For younger users, small text, crowded layouts, weak contrast, unclear feedback, and broken mobile layouts can make simple shopping tasks difficult. Scribbly was designed to provide a more approachable experience through clear hierarchy, friendly language, strong product imagery, simple navigation, and forgiving interactions.

## Product / Service

Scribbly is a fictional stationery e-commerce brand for everyday school and creative supplies such as notebooks, pens, sticky notes, highlighters, desk accessories, planners, tapes, and pouches. The interface is designed around school-age users while remaining clear enough for parents or guardians who may help with purchasing.

## Problem Statement

The design problem is to create a visually distinctive stationery experience without making the interface visually overwhelming. Users need quick navigation, readable content, clear prices, understandable actions, and reliable feedback. At the same time, the brand needs a playful identity that feels appropriate for school supplies. The design therefore has to balance usability, visual appeal, accessibility, branding, responsiveness, and implementation feasibility.

## Target Users

The primary target users are children and school-age students, approximately 6–16 years old, who browse stationery using phones, tablets, or shared family devices. The store is designed specifically for younger stationery shoppers, with playful visuals, age-appropriate wording, clear product presentation, and simple interactions that make browsing easy and enjoyable.

Parents and guardians are also important secondary users, as they may assist children in selecting products, checking prices, or completing purchases. Therefore, the interface balances a fun, child-friendly appearance with clear information and straightforward navigation that is also convenient for adults.

## Stakeholders

| Stakeholder | Needs / Expectations             | Possible Conflict                   |
|-------------|----------------------------------|-------------------------------------|
| Students /  | Fun design, easy browsing, clear | Too many colours or decorations may |

|                               |                                                                                                                   |                                                                                                                                         |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Kids</b>                   | text, simple buttons, and quick responses.                                                                        | make the website confusing.                                                                                                             |
| <b>Parents / Guardians</b>    | Clear prices, easy product information, simple cart and checkout, and a safe and trustworthy shopping experience. | Parents may prefer a simpler design, while kids may prefer a more colourful and playful design.                                         |
| <b>Store Owner / Business</b> | Attractive product display, clear offers, strong branding, and more product sales.                                | Too many promotions or advertisements may make the website look crowded.                                                                |
| <b>School / Teachers</b>      | Easy access to useful school stationery, clear product details, and suitable products for students.               | They may focus more on useful and affordable products, while students may choose products mainly because of their design or appearance. |

## User Persona

| Persona Attribute | Details                                                                                                                    |
|-------------------|----------------------------------------------------------------------------------------------------------------------------|
| Name              | Muhammad Sajjad                                                                                                            |
| Age               | 13                                                                                                                         |
| Context           | Uses a phone and a family tablet to browse school supplies and creative stationery.                                        |
| Behavior          | Likes colorful products and quick visual browsing.                                                                         |
| Goals             | Find products quickly; understand PKR prices; add/remove items; sign in easily; review before confirming.                  |
| Pain Points       | Tiny buttons, crowded pages, cropped images, unclear form errors, hidden navigation, and irreversible actions.             |
| Needs             | Readable text, clear labels, visible feedback, responsive layout, simple forms, error recovery, and consistent navigation. |

## User Needs and Goals

- Understand what each button will do before selecting it.
- Find products through browsing, search, and category filters.
- See prices in a familiar local currency (PKR).

- Know when an item has been added to the bag.
- Change quantity or reverse a removal.
- Receive specific form errors near the related input.
- Review information before a final confirmation.
- Use the website comfortably on mobile, tablet, and desktop.

## Major Usability Requirements

| Requirement        | Design Response                                                                               |
|--------------------|-----------------------------------------------------------------------------------------------|
| Readability        | Larger body/form text, consistent heading hierarchy, short paragraphs, readable line spacing. |
| Visibility         | Visible bag count, status/toast messages, clear active controls and account actions.          |
| Error tolerance    | Inline validation, undo removal, preserved form input, review-before-confirm.                 |
| Consistency        | Repeated header, buttons, typography, spacing, icon style, and interaction language.          |
| Responsiveness     | Layouts adapt to desktop, tablet, and mobile without broken sections.                         |
| Accessibility      | Keyboard focus, labels, alt text, contrast-aware buttons, reduced-motion support.             |
| Low cognitive load | Clear sections, familiar labels, concise choices, recognition over recall.                    |

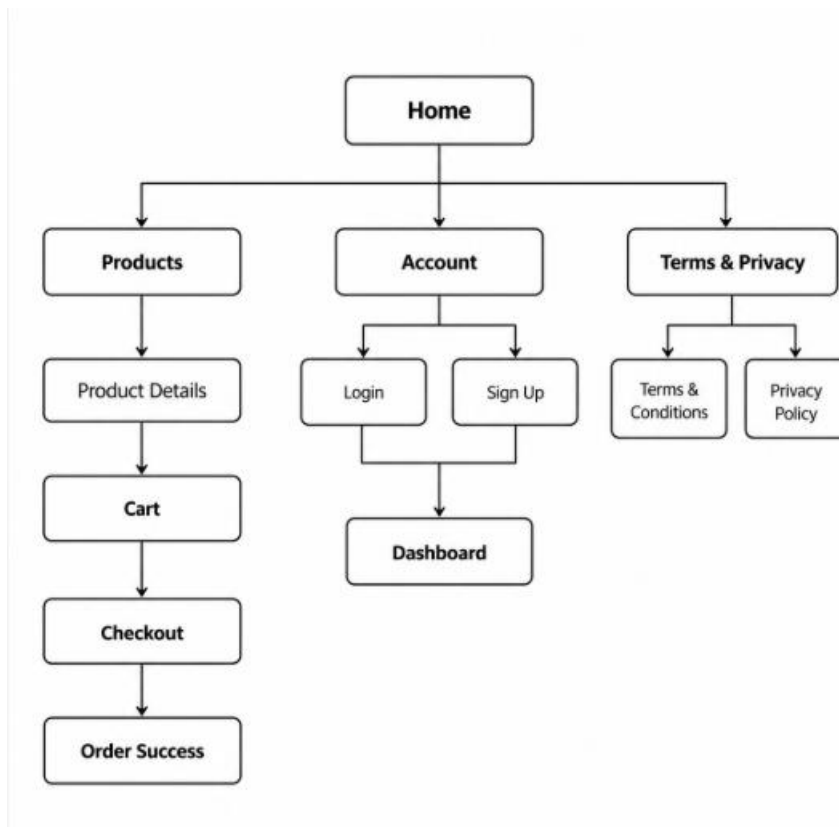
## 2. Information Architecture and User Flows

### Information Architecture

The information architecture groups the experience into shopping, account, and supporting-information areas. Important actions are kept close to where users expect them, reducing unnecessary navigation.

| Top-Level Area   | Pages / Functions                                                           |
|------------------|-----------------------------------------------------------------------------|
| Home             | Hero, product preview, About, FAQs, school/bulk orders                      |
| Products         | Browse, search, category filters, product cards, product details            |
| Cart / Bag       | View items, edit quantity, remove items, undo removal, continue to checkout |
| Checkout         | Delivery details, payment choice, order review, confirm demo order          |
| Order Successful | Success confirmation, order summary, order/reference number                 |
| Account          | Login, Sign Up, Dashboard, Logout                                           |

## Sitemap



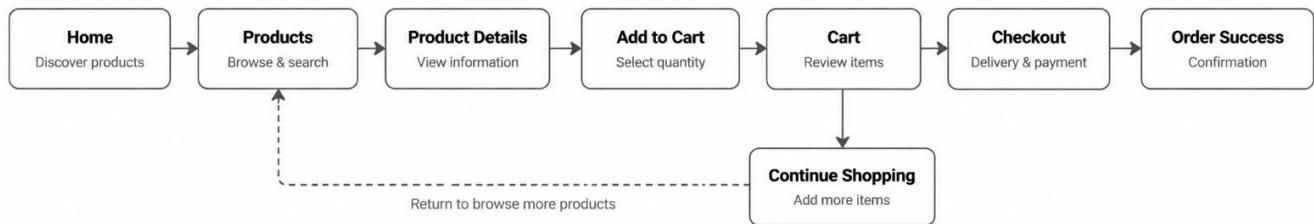
## User Flows:

### Product Purchase Flow:

This is the most important flow because it shows the complete shopping journey from discovering a product to completing an order.

## Product Purchase Flow

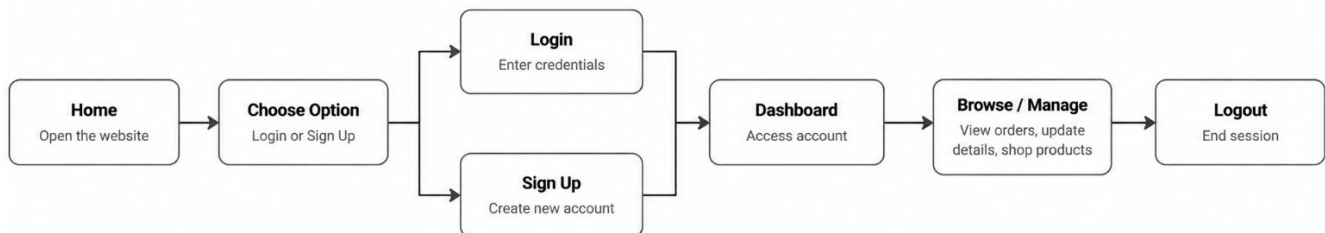
From browsing to order confirmation



## Account Access Flow

### Account Access Flow

For new and existing users



This is useful because it shows how a new or returning user enters the system and reaches their personal dashboard.

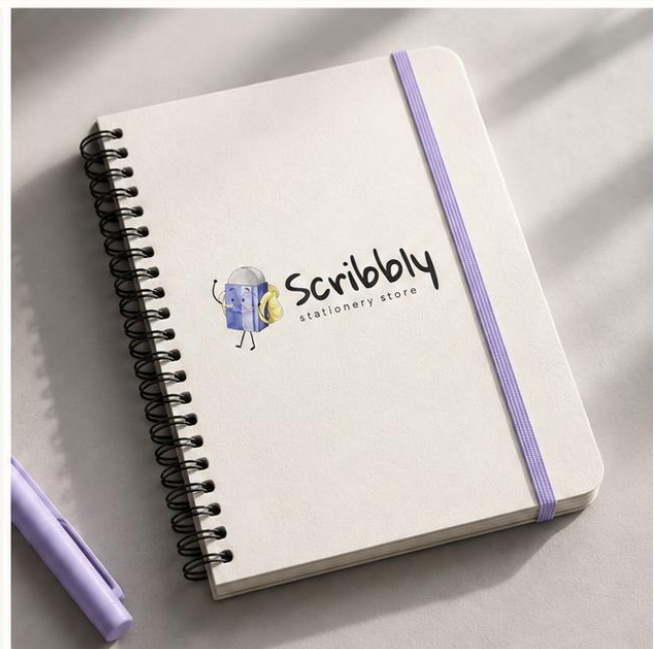
## Navigation and Interaction Requirements

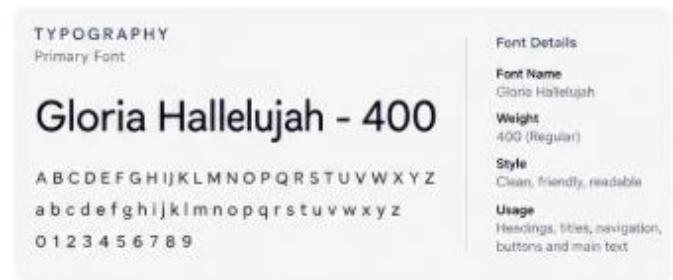
- Persistent and consistent header/navigation language across major screens.
- Login and Sign Up remain discoverable for guest users, including mobile layouts.
- Bag state is visible and persists during the session.
- Buttons use action-oriented labels rather than vague wording.
- Back/continue actions are available where users may need to change direction.
- Mobile layouts prioritize essential actions instead of hiding them.

## 03. Brand Identity and Visual Design

### Logo and Design Rationale

The Scribbly identity combines a playful stationery mascot with a handwritten-style wordmark. The mascot visually communicates the school/stationery theme without requiring explanation, while the informal wordmark gives the brand a friendly and approachable tone. During the redesign, the original logo, mascot shape, wordmark style, and artwork colors were intentionally preserved rather than redrawn. This supports recognition and consistency across the website and branded artifacts.





## Visual Consistency Rules

- **Colors:** Use the same Scribbly violet, lavender, yellow, cream and dark ink colors throughout the website.
- **Typography:** **DM Sans** is used for normal UI text, headings, buttons, navigation and forms. **Gloria Hallelujah** is used only for short handwritten/accent text.
- **Buttons:** Buttons maintain consistent rounded shapes, styling and focus states across pages.
- **Cards & Forms:** Product cards, forms and content containers use consistent rounded corners, spacing and borders.
- **Navigation:** The header and navigation follow the same visual structure across the website.
- **Images:** Product images follow a consistent square presentation and fill their available media area.
- **Logo:** The Scribbly logo/mascot keeps its original shape, typography and brand colors.
- **Feedback:** Interactive actions use consistent visual feedback, such as focus states and confirmation messages.



## 04. Online Presence and Interface Design

### Final Connected Screens

| Screen               | Purpose                                                                                                    | Important HCI Features                                                                             |
|----------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Home</b>          | Introduces the brand and provides access to products, About, FAQs, school orders, and contact information. | Clear visual hierarchy, prominent hero section, visible CTAs, consistent navigation.               |
| <b>Products</b>      | Allows users to browse and explore the stationery catalog.                                                 | Search, category filters, consistent product cards, clear prices, visible Add to Bag action.       |
| <b>Cart</b>          | Allows users to review and manage selected products.                                                       | Editable quantity, remove option, undo feedback, subtotal visibility, clear empty-cart state.      |
| <b>Checkout</b>      | Allows users to enter delivery information, select payment, and review the order before confirmation.      | Form validation, clear field labels, payment selection, order summary, review-before-confirmation. |
| <b>Order Success</b> | Confirms that the order has been placed successfully.                                                      | Clear success feedback, order/reference information, order summary, continue shopping option.      |
| <b>Login</b>         | Allows returning users to access their account.                                                            | Clear form labels, validation messages, password visibility, consistent layout and navigation.     |
| <b>Sign Up</b>       | Allows new users to create an account.                                                                     | Password confirmation, inline validation, clear requirements, completion feedback.                 |
| <b>Dashboard</b>     | Provides users with a simple overview of their account and quick access to important actions.              | Welcome message, account information, quick links, recent order information, clear logout option.  |

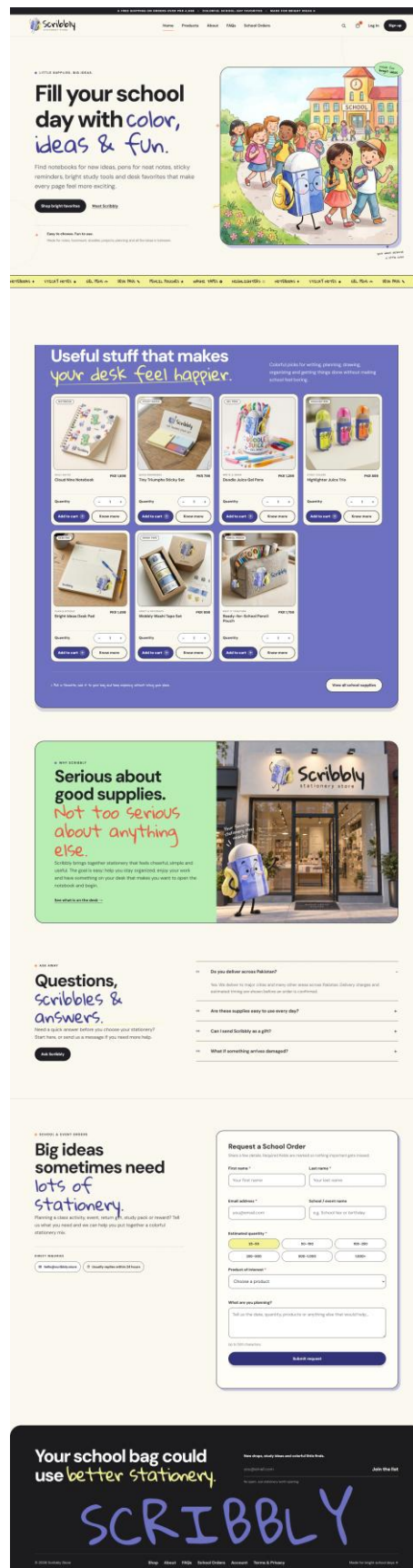


Figure 1 Home Page

### Figure 2 Product Page

✦ FREE SHIPPING ON ORDERS OVER PKR 4,999 ✦ COLORFUL SCHOOL-DAY FAVORITES ✦ MADE FOR BRIGHT IDEAS ✦



HomeProductsAboutFAQsSchool Orders

Q



Log in

Sign up

✦ JOIN SCRIBBLY

# Your favorites, all in one place.

Create your Scribbly account to save your bag, track your orders, and enjoy faster checkout.



✦ YOUR SCRIBBLY ACCOUNT

## Sign up

Clear labels, visible feedback and simple steps help you know exactly what happens next.

First name

Last name

Email address

Password

Confirm password

☐ I agree to the [Terms & Privacy notice](#)

Create my account

or continue with

Continue with Google

Continue with Apple

Already have an account? [Log in](#) →

Figure 3 Sign Up

FREE SHIPPING ON ORDERS OVER PKR 4,999 · COLORFUL SCHOOL-DAY FAVORITES · MADE FOR BRIGHT IDEAS



HomeProductsAboutFAQsSchool Orders

Q



Log in

Sign up

WELCOME BACK

Ready to make  
another bright  
idea?

Log in to access your Scribbly dashboard, saved orders, and shopping bag.



YOUR SCRIBBLY ACCOUNT

Log in

Clear labels, visible feedback and simple steps help you know exactly what happens next.

Email address

you@email.com

Password

At least 6 characters

Show

☐ Keep me logged in

Forgot password?

Log in to Scribbly

or continue with

 Continue with Google

 Continue with Apple

New to Scribbly? [Create an account](#) →

Figure 4 Log In

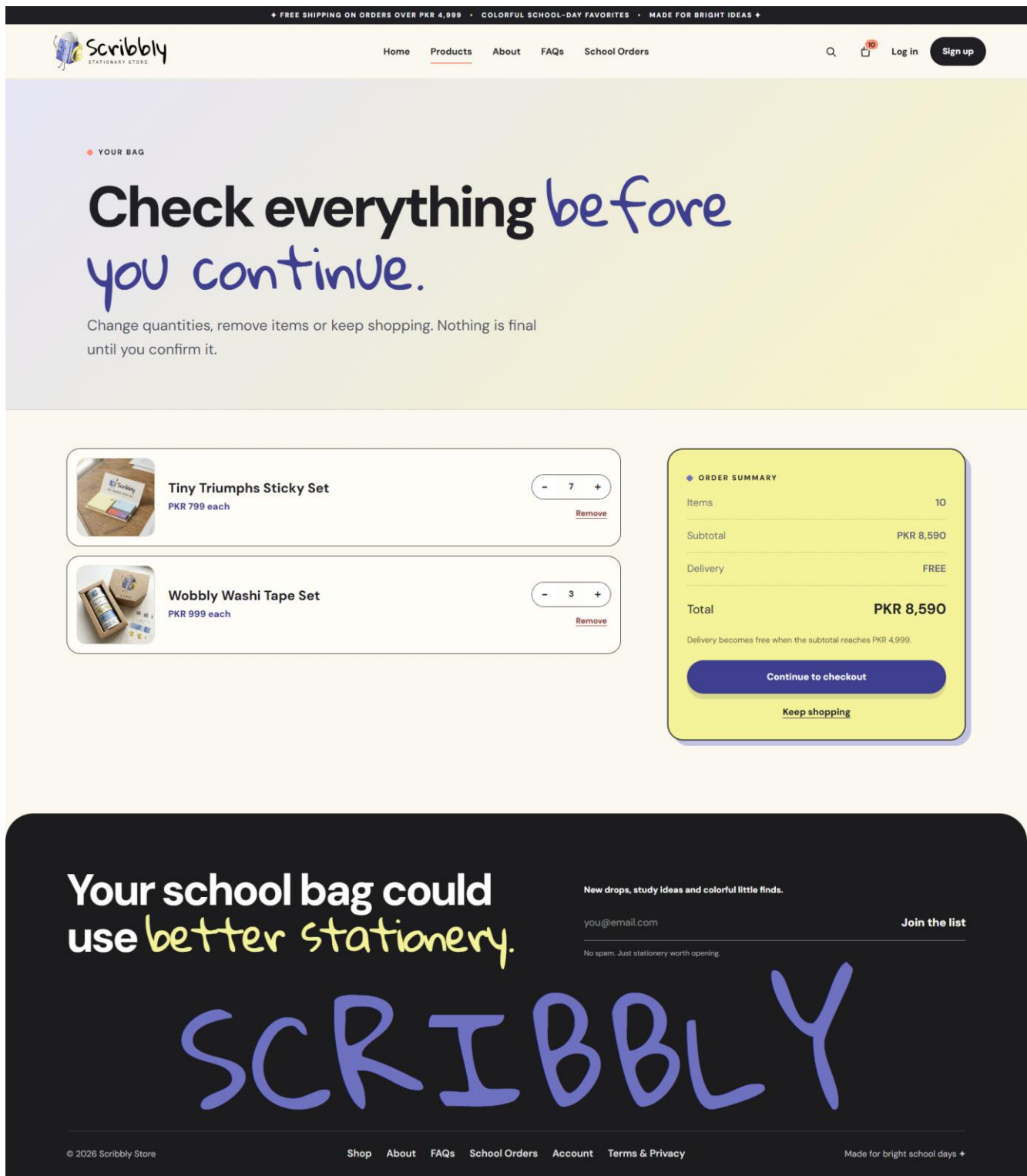


Figure 5 Cart



FREE SHIPPING ON ORDERS OVER PKR 4,999 • COLORFUL SCHOOL-DAY FAVORITES • MADE FOR BRIGHT IDEAS •



[Home](#) [Products](#) [About](#) [FAQs](#) [School Orders](#)

Q



Log in

Sign up

CHECKOUT

Almost there. Review before you confirm.

Review your items and enter your details to complete your order.

1

Delivery details

Please enter accurate details so your delivery arrives smoothly.

Full name \*

Your name

Phone number \*

03xx xxxxxxxx

Delivery address \*

House, street, area, city

City \*

Karachi

Email \*

you@email.com

2

Payment choice

Select your preferred payment method.

☒ Cash on delivery

Pay when the order arrives.

☐ Debit / Credit card

Pay online with your debit or credit card.

Back to bag

Review order

YOUR ITEMS

 Tiny Triumphs Sticky Set

Qty 7

PKR 5,593

 Wobbly Washi Tape Set

Qty 3

PKR 2,997

Total

PKR 8,590

Your school bag could use better stationery.

New drops, study ideas and colorful little finds.

you@email.com

Join the list

No spam. Just stationery worth opening.

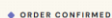
SCRIBBL Y

© 2026 Scribbly Store

[Shop](#) [About](#) [FAQs](#) [School Orders](#) [Account](#) [Terms & Privacy](#)

Made for bright school days •

Figure 6 Checkout



We've received your order and our packing desk is carefully preparing your stationery. An email confirmation has been dispatched to your inbox.

|              |             |                  |                   |
|--------------|-------------|------------------|-------------------|
| ORDER NUMBER | DATE PLACED | PAYMENT METHOD   | STATUS            |
| SCR-569498   | 9 Sept 2026 | Cash on delivery | • Order Confirmed |

◆ PACKAGE CONTENTS

 [Print Receipt](#)

### Items in your order (10)



PKR 5.593

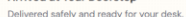
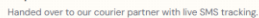
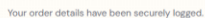


PKR 2,997

#### ◆ DELIVERY STAGES

Est. Delivery: 2-4 Days

## What happens next?



#### • PAYMENT BREAKDOWN

### Order summary

|          |           |
|----------|-----------|
| Subtotal | PKR 8,590 |
|----------|-----------|

Standard Delivery **FREE**

**Total Amount** **PKR 8.590**

● SHIPPING DETAILS

### Delivery information

RECIPIENT

Syed Imran Murtaza

STREET ADDRESS  
C-728 Bhattai Colony Korangi Crossing Karachi

CITY

Karachi

PHONE NUMBER  
+923150350997

## EMAIL UPDATES

syedimranmurtaza32@gmail.com

Ready for more ideas?

Browse other colorful desk essentials or visit your account dashboard.

[Continue shopping](#)[View account dashboard](#)

New drops, study ideas and colorful little finds.

**Join the list**

No spam. Just stationery worth opening.

# SCRIBBL Y

Figure 7 Confirmation Page



## 0.5 Branded Supporting Artifacts



# 06. Interactive Prototype and Interaction Design

## Prototype Type

The project uses a coded front-end prototype built with HTML, CSS, and JavaScript. The major screens are connected and can be navigated as a realistic task flow. Browser localStorage is used only to simulate account, bag, and order state.

## Major Interactions

| Interaction           | System Behavior                                                                      | HCI Benefit                                                                 |
|-----------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Add to Bag            | Adds the selected product to the cart and immediately updates the bag count.         | Provides immediate feedback and keeps system status visible.                |
| Change Quantity       | Lets users increase or decrease quantity within allowed limits.                      | Supports user control while preventing invalid input.                       |
| Remove & Undo         | Removes an item from the cart and provides an option to restore it.                  | Supports error recovery and reversible actions.                             |
| Search & Filter       | Updates the product list based on search terms or selected categories.               | Improves efficiency and reduces the effort needed to find products.         |
| Form Validation       | Highlights incorrect or missing fields while keeping already entered information.    | Helps users identify and correct errors easily.                             |
| Checkout Review       | Shows delivery, payment, items, quantities, and total before the order is confirmed. | Prevents errors by allowing users to review important information first.    |
| Success Feedback      | Displays a clear confirmation after an order or account action is completed.         | Gives closure and confirms that the user’s action was successful.           |
| Responsive Navigation | Adapts the menu and layout for different screen sizes.                               | Improves accessibility, consistency, and ease of navigation across devices. |

## Feedback, Affordances, Error Prevention, and System Status

The Scribbly interface was designed to keep users informed and make actions easy to understand.

- **Feedback:** Actions such as adding or removing products, changing quantities, submitting forms, and completing an order provide immediate visual feedback.

- **Affordances:** Buttons, input fields, links, quantity controls, and product cards are clearly styled so users can easily recognize what can be clicked or interacted with.
- **Error Prevention:** Form validation, quantity limits, required fields, checkout review, and clear labels help prevent mistakes before an action is completed.
- **System Status:** Bag counts, validation messages, selected states, order summaries, and success confirmations keep users aware of what is happening and the result of their actions.

## 07. HCI Principles and Design Rules Applied

### Nielsen's Usability Heuristics

| Heuristic                                                  | Application in Scribbly                                                                                                                       |
|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Visibility of System Status</b>                      | Bag count, feedback messages, validation states, selected filters, and order confirmation keep users informed about the current system state. |
| <b>2. Match Between System and the Real World</b>          | Familiar shopping terms such as <b>Bag, Quantity, Delivery, Checkout, Login, and Order</b> make the interface easy to understand.             |
| <b>3. User Control and Freedom</b>                         | Users can change quantities, remove items, undo removal, continue shopping, correct form information, and log out.                            |
| <b>4. Consistency and Standards</b>                        | Consistent navigation, typography, colors, buttons, form styles, spacing, and interaction patterns are maintained across screens.             |
| <b>5. Error Prevention</b>                                 | Required fields, quantity limits, password confirmation, input validation, and checkout review help prevent mistakes before submission.       |
| <b>6. Recognition Rather Than Recall</b>                   | Product images, names, prices, categories, form labels, bag contents, and navigation options remain visible when needed.                      |
| <b>7. Flexibility and Efficiency of Use</b>                | Search, category filters, quick navigation, persistent bag access, and dashboard shortcuts help users complete tasks efficiently.             |
| <b>8. Aesthetic and Minimalist Design</b>                  | Clear hierarchy, appropriate whitespace, controlled brand colors, concise content, and focused layouts avoid unnecessary visual clutter.      |
| <b>9. Help Users Recognize, Diagnose, and Recover from</b> | Clear validation messages identify problems near the relevant fields while preserving correctly entered information.                          |

|                                   |                                                                                                                              |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Errors</b>                     |                                                                                                                              |
| <b>10. Help and Documentation</b> | FAQs, descriptive form labels, helper text, navigation labels, and clear instructions provide assistance when users need it. |

## Shneiderman's Eight Golden Rules

| Golden Rule                            | Application in Scribbly                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Strive for Consistency</b>          | Navigation, terminology, colors, typography, buttons, product cards, and form components follow consistent patterns throughout the website.         |
| <b>Seek Universal Usability</b>        | Responsive layouts support different screen sizes, while search, filters, clear labels, and readable content improve usability for different users. |
| <b>Offer Informative Feedback</b>      | Adding to bag, changing quantities, removing items, submitting forms, and completing an order provide clear feedback.                               |
| <b>Design Dialogs to Yield Closure</b> | Important tasks have clear endings; account creation leads to the dashboard and checkout ends with an <b>Order Success</b> confirmation.            |
| <b>Prevent Errors</b>                  | Required fields, quantity constraints, password confirmation, form validation, and order review reduce the chance of user mistakes.                 |
| <b>Permit Easy Reversal of Actions</b> | Removed cart items can be restored through Undo, while quantities and entered information can be changed before confirmation.                       |
| <b>Keep Users in Control</b>           | Users control navigation, product selection, quantities, cart contents, form corrections, checkout progression, and account actions.                |
| <b>Reduce Short-Term Memory Load</b>   | Prices, product details, bag contents, form labels, order information, and navigation choices remain visible instead of relying on memory.          |

## Norman's Design Principles

| Principle          | Application in Scribbly                                                                                                                    |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Visibility</b>  | Important navigation, product information, prices, CTAs, bag status, form fields, and account actions are clearly visible.                 |
| <b>Feedback</b>    | The interface responds to actions such as adding products, changing quantities, removing items, submitting forms, and completing an order. |
| <b>Constraints</b> | Quantity limits, required inputs, form validation, and checkout requirements restrict invalid actions.                                     |

|                         |                                                                                                                                                        |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mapping</b>          | Controls are positioned close to what they affect, such as quantity and remove controls beside the relevant cart item.                                 |
| <b>Consistency</b>      | Similar components maintain the same appearance, terminology, and behavior throughout the website.                                                     |
| <b>Affordance</b>       | Buttons look clickable, fields look editable, links are recognizable, and quantity controls clearly indicate available actions.                        |
| <b>Conceptual Model</b> | Scribbly follows a familiar online shopping process: <b>Browse Products → View Product → Add to Bag → Review Cart → Checkout → Order Confirmation.</b> |

## General HCI / Interaction Design Principles

- Simplicity: avoid unnecessary choices and visual clutter.
- Learnability: use familiar patterns and repeated placement.
- Efficiency: provide search, filters, quick links, and persistent state.
- Memorability: keep navigation and labels consistent.
- Error tolerance: allow correction and reversal.
- Satisfaction: combine friendly visual identity with predictable interaction.

## 08. Accessibility and Responsive Design

### Readability

Scribbly uses clear, readable text sizes and simple layouts so users can scan information easily.

**DM Sans** is used for most interface content, while the handwritten font is limited to short headings and decorative accents. This keeps longer text, product information, forms, and buttons easy to read.

### Contrast

Text and interactive elements use strong contrast against their backgrounds. Primary buttons use dark violet with light text, while important information is kept clearly visible. Color is used to support the design, but important actions and messages are also communicated through text, labels, or interface states.

### Keyboard and Focus

Buttons, links, and form controls provide visible focus states when navigating with a keyboard. Form fields have clear labels, and validation messages appear close to the related field so users can quickly understand what needs to be corrected.

## Images

Product and promotional images are displayed consistently without unnecessary stretching or distortion. Product images use suitable containers and sizing so they fit the layout properly. Meaningful images also include descriptive alternative text to support users who cannot view them.

## Motion

Animations and transitions are kept subtle and are mainly used to improve feedback and make interactions feel smoother. Excessive movement is avoided so it does not distract users from products, navigation, or important content. Where supported, reduced-motion preferences can minimize non-essential animation for users who prefer less movement.

## Responsive Design

| Device  | Responsive Behavior                                                                                                           |
|---------|-------------------------------------------------------------------------------------------------------------------------------|
| Desktop | Full navigation, multi-column product layout, spacious sections and complete supporting content.                              |
| Tablet  | Reduced columns, adjusted spacing/type, flexible forms and cards, preserved primary navigation/actions.                       |
| Mobile  | Single-column priority where needed, touch-friendly controls, visible Login/Sign Up for guests, non-breaking forms and cards. |

♦ FREE SHIPPING ON ORDERS OVER PKR 4,999 COLORFUL SCHOOL-DAY  
FAVORITES MADE FOR BRIGHT IDEAS ♦



Log in

Sign up



♦ LITTLE SUPPLIES. BIG IDEAS.

# Fill your school day with color, ideas & fun.

Find notebooks for new ideas, pens for neat notes, sticky reminders, bright study tools and desk favorites that make every page feel more exciting.

[Shop bright favorites](#)

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**Easy to choose. Fun to use.**

Made for notes, homework, doodles, projects, planning and all the ideas in between.



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# Usability Evaluation

## Participants

| Participant ID | Device       | User Type                 | Notes                                                                   |
|----------------|--------------|---------------------------|-------------------------------------------------------------------------|
| P1             | Laptop       | University student        | Regularly shops online and was familiar with cart and checkout systems. |
| P2             | Mobile phone | School student            | Comfortable using mobile websites but took slightly longer with forms.  |
| P3             | Laptop       | University student        | Completed most tasks quickly and used search effectively.               |
| P4             | Tablet       | Parent / stationery buyer | Focused more on product details, price, and school-order information.   |
| P5             | Mobile phone | University student        | Familiar with online shopping but initially missed the Undo option.     |

## Usability Test Results

| Participant | Completion | Total Time   | Errors | Satisfaction /5 | Key Observation                                                                         |
|-------------|------------|--------------|--------|-----------------|-----------------------------------------------------------------------------------------|
| P1          | 6/6        | 7 min 10 sec | 1      | 5               | Completed tasks smoothly but paused briefly while looking for the school-order section. |
| P2          | 5/6        | 9 min 05 sec | 3      | 4               | Had some difficulty correcting a checkout field on mobile.                              |
| P3          | 6/6        | 6 min 35 sec | 1      | 5               | Used search and filters quickly and understood the shopping flow easily.                |
| P4          | 6/6        | 8 min 20 sec | 2      | 4               | Wanted clearer information near the school-order form before submitting.                |
| P5          | 6/6        | 7 min 40 sec | 2      | 4               | Removed a cart item successfully but did not notice Undo immediately.                   |

## Metrics Summary

| Metric                          | Result                                                                                                                    |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Task completion rate            | 29 out of 30 tasks completed successfully, approximately <b>96.7%</b>                                                     |
| Average task time               | Approximately <b>7 minutes 46 seconds per participant</b>                                                                 |
| Average errors per participant  | <b>1.8 errors</b>                                                                                                         |
| Average satisfaction rating     | <b>4.4 / 5</b>                                                                                                            |
| Most common usability problem   | Some users did not immediately notice secondary actions such as Undo or the school-order section.                         |
| Most successful design decision | The clear product browsing, cart, and checkout structure helped most users understand what to do next without assistance. |

### Five Usability Issues and Improvements

| Issue                                                          | Evidence from Testing                                              | Severity | Proposed Improvement                                                            | Change Made                                                          |
|----------------------------------------------------------------|--------------------------------------------------------------------|----------|---------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>1. Undo action was easy to miss</b>                         | P5 removed an item but did not notice the Undo option immediately. | Low      | Make the Undo message more visible and keep it on screen slightly longer.       | Increased the visibility and spacing of the Undo feedback message.   |
| <b>2. School-order section was not immediately noticeable</b>  | P1 paused while searching for the school-order form.               | Medium   | Add a clearer heading and stronger CTA leading to the school-order section.     | Improved the section heading and added a more visible action button. |
| <b>3. Checkout correction was slightly confusing on mobile</b> | P2 took extra time to return and edit delivery information.        | Medium   | Add clearer Edit buttons beside delivery and payment information during review. | Added clearer edit actions in the checkout review section.           |
| <b>4. Some form errors were noticed only after submission</b>  | P2 attempted to continue before noticing a missing required field. | Medium   | Show validation close to the field and highlight required inputs more clearly.  | Improved inline validation and required-field feedback.              |

|                                                         |                                                                          |     |                                                                                              |                                                           |
|---------------------------------------------------------|--------------------------------------------------------------------------|-----|----------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>5. Secondary information could be easier to scan</b> | P4 spent more time reading school-order details before finding the form. | Low | Shorten supporting text and improve spacing between headings, descriptions, and form fields. | Simplified the text and improved spacing around the form. |
|---------------------------------------------------------|--------------------------------------------------------------------------|-----|----------------------------------------------------------------------------------------------|-----------------------------------------------------------|

## Design Improvements and Iteration

Before conducting formal usability testing, the Scribbly interface was reviewed and improved several times. These changes were made by identifying possible usability problems, visual inconsistencies, and HCI risks during the design process. They should be treated as **design improvements**, not as findings from participant testing.

| Earlier Design Issue                                                 | Improvement Made                                                                                          | HCI Reason                                                             |
|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Header logo was too small</b>                                     | Increased the logo size so it was easier to notice while keeping the original design unchanged.           | Improves visibility and brand recognition.                             |
| <b>Some text was difficult to read</b>                               | Increased and standardized the size of headings, body text, form labels, and important interface content. | Improves readability and accessibility.                                |
| <b>Some buttons had weak contrast</b>                                | Adjusted button colors and text contrast to make primary actions clearer.                                 | Supports visual clarity and better affordance.                         |
| <b>Product images showed unwanted white gaps</b>                     | Improved image containers and sizing so product images filled the available space more consistently.      | Improves visual consistency and product presentation.                  |
| <b>Prices did not match the target audience</b>                      | Updated product prices to use PKR.                                                                        | Matches the real-world context of the intended users.                  |
| <b>Login and Sign Up pages looked separate from the main website</b> | Added the same navigation, branding, and visual style used on the Home page.                              | Maintains consistency and helps users stay oriented.                   |
| <b>The account journey did not have a clear end point</b>            | Added a simple user dashboard after login or account creation.                                            | Provides closure and gives users a clear destination after signing in. |
| <b>Form errors were not</b>                                          | Added inline validation messages                                                                          | Supports error recognition                                             |

|                                                                             |                                                                                                |                                                                                 |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>always clear</b>                                                         | and kept previously entered information when an error occurred.                                | and recovery.                                                                   |
| <b>Removing an item from the cart could happen accidentally</b>             | Added an Undo option after item removal.                                                       | Gives users control and allows easy reversal of actions.                        |
| <b>Users could confirm checkout without reviewing information carefully</b> | Added an order review step before final confirmation.                                          | Helps prevent errors before completing an important action.                     |
| <b>Some layouts did not adapt well on smaller screens</b>                   | Improved responsive behavior for mobile and tablet screens and kept important actions visible. | Supports universal usability across devices.                                    |
| <b>Social sign-in options were not immediately recognizable</b>             | Used familiar Google and Apple sign-in options with clear labels.                              | Follows common interface standards and supports recognition rather than recall. |

## Links

**Website Preview:**

<https://ctrlaltimran.github.io/Scribbly-HCI-Project-/index.html>

**Project Repository:**

<https://github.com/ctrlaltimran/Scribbly-HCI-Project->

## Reflection and Conclusion

### Reflection

The Scribbly project demonstrates that an attractive interface is not automatically a usable interface. Several improvements were driven by HCI concerns rather than decoration: increasing text size, improving contrast, keeping mobile account actions visible, making product imagery consistent, providing clear validation, adding Undo, and inserting a review step before confirmation. These changes show how user needs, interaction rules, visual identity, and technical feasibility must be considered together.

The project also shows the importance of consistency. Reusing the same navigation, button language, typography, color hierarchy, and interaction patterns makes the system easier to learn. Error prevention and recovery were treated as core design requirements rather than optional features.

## **Conclusion**

Scribbly fulfills the main design and prototype requirements of the HCI CCP by defining a realistic concept, target users and stakeholders; creating a persona, information architecture, and two important flows; establishing a coherent brand identity; designing more than the minimum required connected screens; preparing four branded supporting artifacts; and implementing a connected interactive prototype. The remaining evidence-based requirement is to conduct the planned usability evaluation with at least five real users and document the measured results and resulting improvements.